Lucas Buffan
Institut des Sciences de l'Évolution de Montpellier

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Dear Editor-in-Chief and Editorial Board members of *Systematic Biology*,

Please find enclosed our manuscript entitled '***A Matter of Size: What Underpinned the Emergence and Demise of the Largest Rodents that Ever Lived?***', which we are pleased to submit for consideration as a research article to *Systematic Biology*.

The factors mediating clade's diversification dynamics as well as the interplay between species diversity and morphological disparity have fascinated evolutionary biologists since the seminal works of George G. Simpson and David M. Raup, more than half a century ago. This study is framed around these cornerstone issues, using Chinchilloidea, a caviomorph rodent clade endemic to the Neotropics, as a case study. Today, Chinchilloidea are poorly diverse, which strikingly contrasts with their rich fossil record over South America, the latter notably encompassing the largest rodents ever recorded. In this work, we investigate the triggers behind the spectacular decline experienced by this clade, in relation to their phenotypic evolution.

To this aim, we built a vetted species-level fossil occurrence dataset, encompassing 708 revised records. In parallel, based on published and new acquisitions of craniodental measurements, we conducted body mass and hypsodonty index reconstructions for 119 out of our 124 species, representing the most exhaustive trait database assembled for this clade. These invaluable data were used to infer the dynamics of chinchilloid diversification, using several distinct cutting-edge occurrence-based frameworks relying on Bayesian modelling and neural networks. These analyses allowed testing for the joint effects of traits, biotic and abiotic variables on chinchilloid diversification. Furthermore, we proposed a novel approach to combine palaeontological and neontological data for diversification inference applied to extant species lacking fossil evidence, using divergence times from dated phylogenies.

Our findings confirm the long-standing paradigm that the temporal dynamics of chinchilloid body size and species richness were coupled. We reveal that the decline of Chinchilloidea was caused by an increase in extinction rate starting in the Late Miocene onwards. This change in extinction regime, selective against large-sized, extratropical taxa, was found to be mediated by the Andean Uplift and sea-level changes, pointing towards an implication of the Late Miocene tectonically- and eustatically-controlled collapse of South American megawetlands.

By providing the most complete trait and occurrence datasets ever assembled for our focal group, our study illustrates how traits and environmental factors synergise to shape the macroevolutionary fate of a clade. Notably, our findings portray how the diversity-disparity relationship, a long-standing research avenue in macroevolution, is impacted by phylogenetic scale. We also evidence a novel mechanistic implication of mountain building for clade's diversification, here enhancing extinction. We also provide an empirical example validating the lower propensity of taxa to go extinct in the tropics relative to outside the tropics, supporting Wallace's 'Tropical Stability' hypothesis.

Our work includes a thorough exploration of the robustness of the methodological approaches implemented to limitations in the input data via well documented sensitivity analyses, and a significant part of the discussion is dedicated to methodological concerns. Also, our study provides new insights into combining neontological and palaeontological data, a milestone in macroevolution. Thus, we think that this manuscript fits the scope of *Systematic Biology*, and may appeal to a broad readership. We thank you for your consideration, and hope that you will share our excitement for the study presented.

Also, for qualified reviewers, we would like to recommend:

- Dr. Luciano L. Rasia (lucianorasia@fcnym.unlp.edu.ar)
- Dr. Juan L. Cantalapiedra (jcantalapiedra@mncn.csic.es)
- Prof. Darin A. Croft (darin.croft@case.edu)
- Dr. Renan Maestri (renanmaestri@gmail.com)
- Prof. Lee Hsiang Liow (l.h.liow@nhm.uio.no)

Lucas Buffan, on behalf of my co-authors